TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**BEAST – Embedded Medical Secure Platform for End-to-End Data Security from SoC to Cloud**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (.), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information Security Technology		Bio and Medical Devices	
	Medium Category	Small Category	Medium Category	Small Category
	Embedded Security Systems	Secure Medical IoT Platforms	Digital Healthcare Systems	Secure Medical Data Acquisition and Transmission
Concept of Proposed Technology	There is an increasing number of **medical IoT devices** exchanging sensitive personal data with multiple stakeholders over **unsecured networks and storage infrastructures**, creating major **security and confidentiality challenges**. The proposed technology, named **BEAST (Embedded Medical Secure Platform)**, is an **end-to-end secure communication and data exchange platform**, providing **security and confidentiality from the sensor node up to the cloud**, based on **commercial off-the-shelf (COTS) hardware components**. BEAST has been developed by CSEM and is based on the **Nordic nRF5340 SoC**, leveraging CSEM expertise in **embedded security** and **PPG (Photoplethysmography) acquisition and processing**. The platform is particularly suited for applications acquiring and exchanging **sensitive medical and healthcare data**. BEAST implements: • a **custom transport layer over Bluetooth Low Energy (BLE)**, • a **security protocol and key management architecture**, • and a **complete secure channel establishment mechanism** across all stakeholders. •			
Rationale for Joint R&D Project	BLE-based medical devices typically rely on **standard pairing procedures**, which are cumbersome and impractical in environments with: • sensitive data, • many dynamically changing users (e.g. hospitals), • and strict usability constraints. Furthermore, security is required at two levels:			

	1. Ensuring that sensitive data is transmitted only to a **trusted infrastructure** (device → gateway → cloud). 2. Preventing **third-party applications** from intercepting medical data when using mobile or gateway-based access. The BEAST platform addresses these issues by providing an **automated, device-trust-based secure communication framework**. Joint R&D is required to: • adapt and validate the platform with Korean SME medical IoT products, • integrate it into real deployment environments, • and prepare it for industrial-scale use.
Key Words	1. Medical IoT security 2. End-to-end secure communication 3. Bluetooth Low Energy (BLE) 4. Hardware-based trust 5. Secure data exchange

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5-6
R&D Objectives	• To provide an easy-to-deploy end-to-end secure platform for medical IoT devices. • To secure data acquisition, transmission, and exchange from sensor node to cloud. • To implement automated secure channel establishment without user-driven pairing. • To ensure confidentiality, integrity, and authentication across all system components.			
R&D Target Specifications	• Mutual authentication based on **SIGMA-R protocol.** • Secure session establishment using **AEAD** • Support for BLE-based secure payload transmission • Secure storage and management of device and infrastructure certificates • Compatibility with **TF-M compliant MCUs**			
R&D Plan	**Phase 1 – Security Architecture and Protocol Integration** • Integration of custom BLE transport layer • Implementation of SIGMA-R–based mutual authentication • Definition of PKI hierarchy and certificate management **Phase 2 – Embedded and Gateway Implementation** • End node implementation on nRF5340 using Zephyr RTOS and TF-M • Gateway implementation on Linux (Raspberry Pi 5) • BLE communication using Zephyr BLE stack and BlueZ **Phase 3 – System Validation** • Secure channel establishment validation • Transmission of medical PPG and heart rate data • End-to-end secure data visualization via gateway and cloud interface			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for: • Design and implementation of the BEAST security architecture • Development of embedded security software based on TF-M • Definition and management of PKI hierarchy and security protocols • System integration and validation
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: • Provision of medical IoT use cases and system requirements • Integration of BEAST into medical devices or gateways • Testing and validation in product-relevant environments • Contribution to deployment and scalability considerations

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will result in: • a validated **end-to-end secure medical IoT platform**, • simplified deployment of secure BLE-based medical devices, • and improved confidentiality and integrity of sensitive healthcare data. For Korean SMEs, the impact includes: • faster development of compliant medical IoT products, • reduced security integration complexity, • and enhanced trustworthiness in healthcare data exchange ecosystems.

6. References

(1) BEAST – Embedded Medical Secure Platform: Securing Data from the SoC up to the Cloud

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization